Session – 18

Task-1.

Write an SQL query to display the total number of songs uploaded by each artist from a table 'songs' (columns: song_id, artist_name, title) and show only those artists who have uploaded more than 3 songs.

Solution:

```
SELECT artist_name, COUNT(*) AS total_songs
FROM songs
GROUP BY artist_name
HAVING COUNT(*) > 3;
```

Task-2.

Given two tables, 'orders' (order_id, user_id, amount) and 'users' (user_id, username), write a SQL JOIN query to display each username along with their total order amount.

Solution:

```
SELECT u.username, SUM(o.amount) AS total_order_amount
FROM users u
JOIN orders o
  ON u.user_id = o.user_id
GROUP BY u.user_id, u.username;
```

Task-3.

Write a SQL subquery to find the names of all restaurants from a 'restaurants' table (id, name, rating) whose rating is higher than the average rating of all restaurants.

Solution:

```
SELECT name, rating
FROM restaurants
WHERE rating > (
  SELECT AVG(rating)
  FROM restaurants
);
```

Task-4.

Using a 'transactions' table (id, user_id, amount, transaction_date), write a SQL query with a window function to display each user's transaction amount and their running total (cumulative sum) ordered by transaction_date.

Solution:

```
SELECT
  user_id,
  transaction_date,
  amount,
  SUM(amount) OVER (
    PARTITION BY user_id
    ORDER BY transaction_date
  ) AS running_total
FROM transactions;
```

Task-5.

List two optimizations you would apply to speed up a query that filters Flipkart products by category and price, and briefly explain how each helps.

Hint: Think about indexes and query structure.

Solution:

```
SELECT product_id, product_name, price
FROM products
WHERE category = 'Electronics'
 AND price < 50000;
```